

19 Augustus/*August 2003*Van/*Surname* _____ Voorletters/*Initials* _____Stud. no./ *Stud. nr.* _____ Handtekening/*Signature* _____

Maksimum punte: 100

Maximum marks: 100

Tyd: 120 minute

Time: 120 minutes

1 Beantwoord alle vrae op die vraestel.

Answer all questions in this paper.

2 Toon alle berekenings volledig aan.

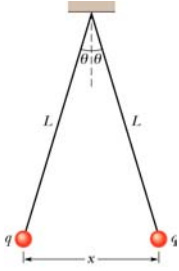
Show all calculations in detail.

3 Die gebruik van programmeerbare sakrekenaars is nie toelaatbaar.

The use of programmable calculators is prohibited.

4 Slegs die handboeke gebruik in hierdie kurses is toelaatbaar.

*Only the text books used in this course are allowed.***Voltooi asseblief die gedeelte hieronder, skeur dit af en handig in voordat met die toets begin word.*****Please complete the section below, tear it off and hand in before commencing with test.***-----
FSK 126 – SEMESTERTOETS/ *TEST* 1**19 Augustus/*August 2003***Van/*Surname* _____ Voorletters/*Initials* _____Stud. no./ *Stud. nr.* _____ Handtekening/*Signature* _____

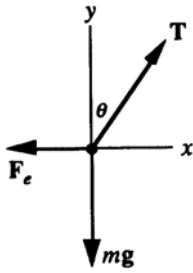


1. In die figuur, hang twee klein geleidende balle met identiese massa m and identiese lading q van nie geleidende drade met lengte L . Neem aan dat θ so klein is dat $\tan \theta$ vervang kan word met sy ongeveer gelyke, $\sin \theta$. Toon dat vir ewewig, $x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$, waar x die afstand tussen die balle is.

In the figure, two tiny conducting balls of identical mass m and identical charge q hang from nonconducting threads of length L . Assume that θ is so small that $\tan \theta$ can be replaced by its

approximate equal, $\sin \theta$. Show that, for equilibrium, $x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$, where x is the separation between the balls.

(10)



$$\sum F_y : T \cos \theta - mg = 0$$

$$T = \frac{mg}{\cos \theta}$$

$$\sum F_x : T \sin \theta - F_e = 0$$

$$mg \tan \theta - F_e = 0$$

$$\sin \theta = \frac{x}{2L}$$

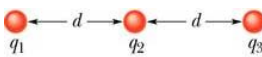
$$F_e = \frac{q^2}{4\pi\epsilon_0 x^2}$$

$$\therefore mg \frac{x}{2L} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2}$$

$$x^3 = \frac{1}{4\pi\epsilon_0} \frac{2q^2 L}{mg}$$

$$x = \left(\frac{1}{2\pi\epsilon_0} \frac{q^2 L}{mg} \right)^{1/3} \text{ m}$$

2. Die drie ladings in die figuur lê in 'n reguit lyn en is 'n afstand d uitmekaar. Ladings q_1 en q_2 is vas terwyl lading q_3 vry is om te beweeg maar is in ekwilibrium (geen netto krag werk daarop). Bepaal q_1 in terme van q_2 .



In the figure, three charged particles lie on a straight line and are separated by distances d . Charges q_1 and q_2 are held fixed. Charge q_3 is free to move but happens to be in equilibrium (no net electrostatic force on it). Find q_1 in terms of q_2 .

(4)

$$k \frac{q_1 q_3}{(2d)^2} + k \frac{q_2 q_3}{d^2} = 0$$

$$\frac{q_1}{4} + q_2 = 0$$

$$q_1 = -4q_2$$

3. Die ladings en koördinate van twee gelaaide deeltjies wat vas gehou word in die xy vlak is: $q_1=3.0 \mu\text{C}$, $x_1=4.0 \text{ cm}$, $y_1=1.0 \text{ cm}$ en $q_2=-4 \mu\text{C}$, $x_2=-2.0 \text{ cm}$, $y_2=1.5 \text{ cm}$. Bepaal die grootte en rigting van die elektrostatiese krag op q_2 .

The charges and coordinates of two charged particles held fixed in the xy plane are: $q_1=3.0 \mu\text{C}$, $x_1=4.0 \text{ cm}$, $y_1=1.0 \text{ cm}$ and $q_2=-4 \mu\text{C}$, $x_2=-2.0 \text{ cm}$, $y_2=1.5 \text{ cm}$. Find the magnitude and direction of the electrostatic force on q_2 . (5)

$$F_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \frac{(8.99 \times 10^9)(3 \times 10^{-6})(4 \times 10^{-6})}{[(-2 - 4.0)^2 + (1.5 - 1.0)^2] \times 10^{-4}}$$

$$= 29.8 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \tan^{-1} \left(\frac{0.5}{-6.0} \right) = -4.8^\circ$$

Waar sal 'n derde lading $q_3=4.0 \mu\text{C}$ geplaas moet word sodat die netto elektrostatiese krag op q_2 nul is?

Where must you locate a third charge $q_3=4.0 \mu\text{C}$ such that the net electrostatic force on q_2 is zero? (5)

Vir $\sum F = 0 \Rightarrow q_3$ op dieselfde lyn as q_1 en q_2

$$F_{23} = \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r^2} = F_{21}$$

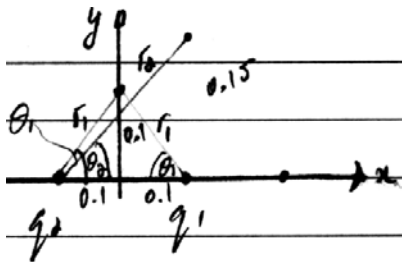
$$\therefore r = \sqrt{\frac{q_2 q_3}{4\pi\epsilon_0 F_{21}}} = \sqrt{\frac{(4 \times 10^{-6})(4 \times 10^{-6})(8.99 \times 10^9)}{29.8}} = 6.9 \text{ cm}$$

$$x_3 = x_2 - r \cos \theta = -2 - (6.9 \cos(-4.8)) = -8.9 \text{ cm}$$

$$y_3 = y_2 - r \sin \theta = 1.5 - (6.9 \sin(-4.8)) = 2.1 \text{ cm}$$

4. 'n Positiewe punt lading met 'n grootte van $4.00 \times 10^{-8} \text{ C}$ word in 'n reghoekige koördinaat sisteem by 'n punt $x = +0.10 \text{ m}$, $y = 0$ geplaas. 'n Identiese punt lading word by 'n punt $x = -0.10 \text{ m}$, $y = 0$ geplaas. Bepaal die grootte en rigting van die elektriese veld by die oorsprong.

In a rectangular coordinate system a positive point charge with a magnitude of $4.00 \times 10^{-8} \text{ C}$ is placed at a point $x = +0.10 \text{ m}$, $y = 0$. An identical point charge is placed at $x = -0.1 \text{ m}$, $y = 0$. Calculate the magnitude and direction of the electric field at the origin. (4)



$$E_1 = E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$= \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.1)^2}$$

$$= 3.6 \times 10^4 \text{ N/C}$$

$$\text{maar } E_1 = -E_2$$

$$\therefore E = 0$$

Bepaal die grootte en rigting van die elektriese veld by die punt $x = 0.20 \text{ m}$, $y = 0$.

Calculate the magnitude and direction of the electric field at the point $x = 0.20 \text{ m}$, $y = 0$.

(5)

$$E_1 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.1)^2}$$

$$= 3.6 \times 10^4 \text{ N/C}$$

$$E_2 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.3)^2}$$

$$= 4 \times 10^3 \text{ N/C}$$

$$E = E_1 + E_2$$

$$= 3.6 \times 10^4 + 4 \times 10^3$$

$$= 4 \times 10^4 \text{ N/C}$$

$$\theta = 0^\circ$$

Bepaal die grootte en rigting van die elektriese veld by die punt $x = 0.10 \text{ m}$, $y = 0.15 \text{ m}$.

Calculate the magnitude and direction of the electric field at the point $x = 0.10 \text{ m}$, $y = 0.15 \text{ m}$.

(9)

$$E_1 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.15)^2} = 1.6 \times 10^4 \text{ N/C}$$

$$r_2 = \sqrt{0.15^2 + 0.2^2} = 0.25 \text{ m}$$

$$E_2 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.25)^2} = 5.76 \times 10^3 \text{ N/C}$$

$$\theta_2 = \sin^{-1}\left(\frac{0.15}{0.25}\right) = 36.87^\circ$$

$$E_{2x} = 5.76 \times 10^3 \cos 36.87^\circ$$

$$= 4.61 \times 10^3 \text{ N/C}$$

$$E_{2y} = 5.76 \times 10^3 \sin 36.87^\circ$$

$$= 3.46 \times 10^3 \text{ N/C}$$

$$\vec{E} = (E_{2x})\mathbf{i} + (E_{1y} + E_{2y})\mathbf{j}$$

$$= (4.61 \times 10^3 \mathbf{i} + 1.95 \times 10^4 \mathbf{j}) \text{ N/C}$$

$$E = \sqrt{(4.61 \times 10^3)^2 + (1.95 \times 10^4)^2}$$

$$= 2 \times 10^4 \text{ N/C}$$

$$\text{rigting} = \tan^{-1}\left(\frac{1.95}{0.46}\right)$$

$$= 76.67^\circ$$

Bepaal die grootte en rigting van die elektriese veld by die punt $x = 0, y = 0.10 \text{ m}$.

Calculate the magnitude and direction of the electric field at the point $x = 0, y = 0.10 \text{ m}$.

(7)

$$r_1 = \sqrt{(0.1)^2 + (0.1)^2} = 0.141 \text{ m}$$

$$\theta_1 = \tan^{-1}\left(\frac{0.1}{0.1}\right) = 45^\circ$$

$$E_1 = E_2 = \frac{(8.99 \times 10^9)(4 \times 10^{-8})}{(0.141)^2}$$

$$= 1.81 \times 10^4 \text{ N/C}$$

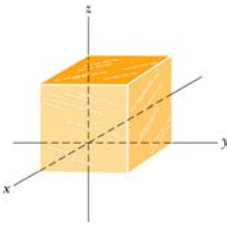
simmetrie $\Rightarrow x$ – komp^e kanseleer

$$\therefore E = 2E_y$$

$$= 2(1.81 \times 10^4) \sin 45^\circ$$

$$= 2.56 \times 10^4$$

$$\text{rigting} = 90^\circ$$



5. By elke punt op die oppervlak van die kubus is die elektriese veld in die positiewe z -as rigting. Die lengte van elke sy van die kubus is 3.0 m . By die boonste oppervlak van die kubus is $\vec{E} = -34 \hat{k} \text{ N/C}$ en op die onderste oppervlak van die kubus is $\vec{E} = +20 \hat{k} \text{ N/C}$. Bepaal die netto lading bevat deur die kubus?

At each point on the surface of the cube, the electric field is in the positive direction of z . The length of each edge of the cube is 3.0 m . On the top surface of the cube $\vec{E} = -34 \hat{k} \text{ N/C}$, and on

the bottom face of the cube $\vec{E} = +20 \hat{k} \text{ N/C}$. Determine the net charge contained within the cube.

(6)

for front, back, left and right surfaces

$$\int \vec{E} \cdot d\vec{A} = 0 \text{ as the angle between the vectors} = 90^\circ$$

for top and bottom surfaces

$$\Phi = \int_{\text{top}} \vec{E} \cdot d\vec{A} + \int_{\text{bottom}} \vec{E} \cdot d\vec{A}$$

$$= \int -34 \hat{k} \cdot dA \hat{k} + \int 20 \hat{k} \cdot dA (-\hat{k})$$

$$= -34(3.0)^2 - 20(3.0)^2$$

$$= -486 \text{ Nm}^2 / \text{C}$$

$$q_{\text{enc}} = \epsilon_0 \Phi$$

$$= 8.85 \times 10^{-12} (-486)$$

$$= -4.3 \times 10^{-9} \text{ C}$$

6. Twee lang, gelaaide, konsentriese silinders het radiusse van 3.0 en 6.0 cm. Die lading per eenheidslengte is 10.0×10^{-6} C/m op die binneste en -12.0×10^{-6} C/m op die buitenste silinder. Bepaal die elektriese veld by $r = 4.0$ cm.
Two long, charged, concentric cylinders have radii of 3.0 and 6.0 cm. The charge per unit length is 10.0×10^{-6} C/m on the inner and -12.0×10^{-6} C/m on the outer cylinder. Find the electric field at $r = 4.0$ cm. (3)

$$E = \frac{\lambda_i}{2\pi\epsilon_0 r} = \frac{10 \times 10^{-6}}{2\pi(8.85 \times 10^{-12})(4 \times 10^{-2})}$$

$$= 4.50 \times 10^6 \text{ N/C}$$

Bepaal die elektriese veld by $r = 8.0$ cm.

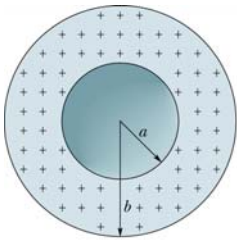
Find the electric field at $r = 8.0$ cm

(3)

$$E = \frac{\lambda_i + \lambda_o}{2\pi\epsilon_0 r}$$

$$= \frac{(10 \times 10^{-6}) - (12 \times 10^{-6})}{2\pi(8.85 \times 10^{-12})(8 \times 10^{-2})}$$

$$= -4.5 \times 10^5 \text{ N/C}$$



7. Die figuur wys 'n sferiese ladings skil met univorme volume ladings digtheid ρ . Bepaal uitdrukkings vir E a.g.v die skil vir afstande r vanaf die middelpunt van die skil tot 30 cm en skets E kwalitatiewelik. Neem aan dat $\rho = 1.0 \times 10^{-6}$ C/m³, $a = 10$ cm, en $b = 20$ cm.

The figure shows a spherical shell of charge with uniform volume charge density ρ . Determine expressions for E due to the shell for distances r from the center of the shell ranging from zero to 30 cm and make a qualitatively plot of E . Assume that $\rho = 1.0 \times 10^{-6}$ C/m³, $a = 10$ cm, and $b = 20$ cm. (10)

For $r < a$: $E = 0$

For $a < r < b$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi(r^3 - a^3)\rho}{r^2}$$

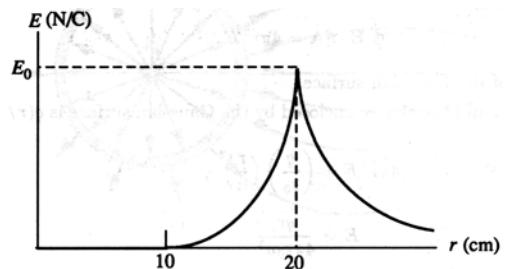
$$= \frac{\rho}{3\epsilon_0 r^2} (r^3 - a^3)$$

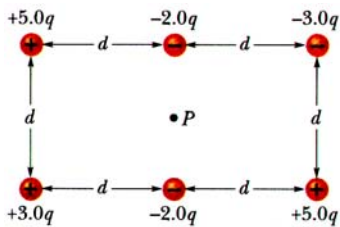
For $r > b$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\frac{4}{3}\pi(b^3 - a^3)\rho}{r^2}$$

$$= \frac{\rho}{3\epsilon_0 r^2} (b^3 - a^3)$$





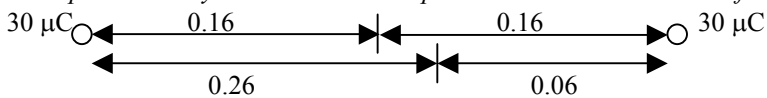
8. Punt P is by die middel van die reghoek. Wat is die netto elektriese potensiaal by P a.g.v. die ses ladings?

Point P is at the centre of the rectangle. What is the net electric potential at P due to the six charges? (8)

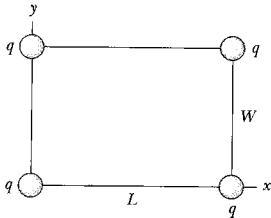
$$\begin{aligned}
 V &= \sum_{i=1}^6 V_{Pi} = \sum_{i=1}^6 \frac{q_i}{4\pi \epsilon_0 r_i} \\
 &= \frac{1}{4\pi \epsilon_0} \left[\frac{5q}{\sqrt{d^2 + (d/2)^2}} + \frac{-2q}{d/2} + \frac{-3q}{\sqrt{d^2 + (d/2)^2}} + \frac{+3q}{\sqrt{d^2 + (d/2)^2}} + \frac{-2q}{d/2} + \frac{+5q}{\sqrt{d^2 + (d/2)^2}} \right] \\
 &= \frac{1}{4\pi \epsilon_0} \left[\frac{5q}{\frac{\sqrt{5}}{2}d} - \frac{2q}{\frac{d}{2}} - \frac{3q}{\frac{\sqrt{5}}{2}d} + \frac{3q}{\frac{\sqrt{5}}{2}d} - \frac{2q}{\frac{d}{2}} + \frac{5q}{\frac{\sqrt{5}}{2}d} \right] \\
 &= \frac{1}{4\pi \epsilon_0} \left[\frac{10q}{\sqrt{5}d} - \frac{4q}{d} - \frac{6q}{\sqrt{5}d} + \frac{6q}{\sqrt{5}d} - \frac{4q}{d} + \frac{10q}{\sqrt{5}d} \right] = \frac{0.94q}{4\pi \epsilon_0 d}
 \end{aligned}$$

9. Twee $30 \mu\text{C}$ ladings word 32 cm van mekaar geplaas. Hoeveel werk word vereis om 'n $0.5 \mu\text{C}$ toets lading van 'n punt halfpad tussen hulle na 'n punt 10 cm nader aan enige van hulle te beweeg?

Two $30 \mu\text{C}$ charges are placed 32 cm apart from each other. How much work is required to move a $0.5 \mu\text{C}$ test charge from a point midway between them to a point 10 cm closer to either of them? (8)



$$\begin{aligned}
 V_1 &= \frac{kQ}{r_1} + \frac{kQ}{r_2} \\
 &= \frac{2kQ}{0.16} \\
 &= \frac{2(8.99 \times 10^9)(30 \times 10^{-6})}{0.16} \\
 &= 3.37 \times 10^6 \text{ V} \\
 V_2 &= \frac{kQ}{0.26} + \frac{kQ}{0.06} \\
 &= \frac{(8.99 \times 10^9)(30 \times 10^{-6})}{0.26} + \frac{(8.99 \times 10^9)(30 \times 10^{-6})}{0.06} \\
 &= 5.53 \times 10^6 \text{ V} \\
 \Delta V &= V_2 - V_1 \\
 &= (5.53 - 3.37) \times 10^6 \\
 &= 2.16 \times 10^6 \text{ V} \\
 W &= q\Delta V \\
 &= (0.5 \times 10^{-6})(2.16 \times 10^6) \\
 &= 1.08 \text{ J}
 \end{aligned}$$



10. Vier identiese punt ladinge ($q = +10.0 \mu\text{C}$) is op die vier hoeke van 'n reghoek geplaas soos aangetoon. Die afmetings van die reghoek is $L = 60.0 \text{ cm}$ en $W = 15.0 \text{ cm}$. Bereken die elektriese potensieële energie van die lading op die onderste, linkerkantste hoek as gevolg van die ander ladinge.

Four identical point charges ($q = +10.0 \mu\text{C}$) are located on the corners of a rectangle as shown. The dimensions of the rectangle are $L = 60.0 \text{ cm}$ and $W = 15.0 \text{ cm}$. Calculate the electric potential energy of the charge at the lower left corner due to the other charges. (5)

$$\begin{aligned}
 U_e &= q_4V_1 + q_4V_2 + q_4V_3 = q_4 \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \right) \\
 &= (10.0 \times 10^{-6})(8.99 \times 10^9) \left(\frac{10.0 \times 10^{-6}}{0.60} + \frac{10.0 \times 10^{-6}}{0.15} + \frac{10.0 \times 10^{-6}}{\sqrt{(0.60)^2 + (0.15)^2}} \right) \\
 &= 8.95 \text{ J}
 \end{aligned}$$

11. Wat is die oortollige lading op 'n geleidende sfeer met radius $r = 0.15 \text{ m}$ as die potensiaal van die sfeer 1500 V is en $V = 0$ by oneindig?

What is the excess charge on a conducting sphere of radius $r = 0.15 \text{ m}$ if the potential of the sphere is 1500 V and $V = 0$ at infinity? (3)

$$\begin{aligned}
 V &= q / 4\pi\epsilon_0 r \\
 q &= 4\pi\epsilon_0 r V = \frac{(0.15)(1500)}{8.99 \times 10^9} \\
 &= 2.5 \times 10^{-8} \text{ C}
 \end{aligned}$$

12. Twee elektrone is vas 2.0 cm van mekaar. 'n Derde elektron word vanaf oneindig geskiet en stop halfpad tussen die eerste twee. Wat is die begin snelheid van die derde elektron?

Two electrons are fixed 2.0 cm apart. Another electron is shot from infinity and stops halfway between them. What is the initial speed of the third electron? (5)

$$\begin{aligned}
 U_f &= \frac{1}{4\pi\epsilon_0} \frac{e^2}{d} + \frac{1}{4\pi\epsilon_0} \frac{e^2}{d} \\
 &= \frac{2}{4\pi\epsilon_0} \frac{e^2}{d} \\
 U_f &= K_i \\
 \frac{2}{4\pi\epsilon_0} \frac{e^2}{d} &= \frac{1}{2} mv^2 \\
 v &= \sqrt{\frac{4e^2}{4\pi\epsilon_0 dm}} \\
 &= \sqrt{\frac{(8.99 \times 10^9)(4)(1.6 \times 10^{-19})^2}{0.01(9.11 \times 10^{-31})}} \\
 &= 3.2 \times 10^2 \text{ m/s}
 \end{aligned}$$